3D Slicer Bounding Box Navigator

User Guide

Note: Please install the latest version of 3D Slicer before you start.

Quick Setup

1. **Download the ZIP** from `github.com/PieterGort/3d-slicer-annotation-tool`: **Code** → **Download ZIP**.
2. **Extract the ZIP.** On Windows and Linux, extract the zip file in Downloads. On macOS, extract the zip file in your home folder (Finder: Cmd+Shift+H). Don't extract in Downloads, Desktop, or Documents, or the module disappears after a restart.
3. **Open the launch file.**
   - **Windows:** double-click `launch_windows.bat`.
   - **Linux:** run `./launch_linux.sh`.
   - **macOS:** double-click `launch_mac.command`. If macOS blocks it, right-click it, choose **Open**, and click **Open** in the warning dialog.

Manual Setup

If the launch file did not open the module, add it inside Slicer once:

1. **Download the ZIP** from `github.com/PieterGort/3d-slicer-annotation-tool`: **Code** → **Download ZIP**.
2. **Extract the ZIP.** On Windows and Linux, extract the zip file in Downloads. On macOS, extract the zip file in your home folder (Finder: Cmd+Shift+H). Don't extract in Downloads, Desktop, or Documents, or the module disappears after a restart.
3. Open **3D Slicer**.
4. In the menu, go to **Edit** → **Application Settings** → **Modules**.
5. Under **Additional module paths**, click **Add** and select the **`BoundingBoxNavigator`** folder.
6. Click **OK** and restart Slicer.

How to Annotate

1. **Open the Module**
   - In the top toolbar, go to: **Modules** → **Annotation** → **Bounding Box Navigator** to display the panel on the right.
2. **Select Folders & Scan**
   - Set **Input Scans Folder** (where your CT scans are) and **Output Annotations Folder**.
   - Click **Scan Folder**. The first unreviewed case loads automatically.
   - Each slice view shows the **original** scan for that plane when the file exists; otherwise it reconstructs from the axial volume (a warning is shown).
3. **Draw Bounding Boxes**
   - Choose a **Finding type** (default: PM nodule). Then press **B** (or click **+ Add Bounding Box**).
   - **Click and drag** across the lesion on any slice view (axial, coronal, or sagittal).
   - Drag the colored handles on the box to resize it in width, height, and depth.
   - Scroll through slices with your mouse wheel to check coverage.
   - Use the large **slice thickness** buttons (3 mm, TS, . . .). For TS (or any thickness without native COR/SAG), coronal and sagittal show reconstructions of that axial scan.
   - Press **B** again for each additional lesion.
   - *To hide or show a box*: untick / tick the **Show** checkbox in front of it in the box list. Hidden boxes are still saved; this only hides them while you look at the scan.
   - *To remove a box*: click it in the list and press **Delete Selected**.
4. **Notes / PCI Score**
   - Type the PCI score or clinical comments in the **Notes / PCI** field (optional).
   - Optional: in **4. Segmentation**, load a `.seg.nrrd` or `.nii.gz` overlay (`.nii.gz` is imported as a labelmap, not a CT volume).
5. **Save and Next Scan**
   - Click **Save + Next** (large green button).
   - Each box is saved as `R_1.mrk.json`, `A_1.mrk.json`, . . . inside a folder named after the case ID.
   - The next unreviewed scan loads immediately.
   - *If there are 0 lesions (PCI 0)*: click **Save + Next** and confirm with **Yes**.

Mouse & Keyboard Shortcuts

Shortcut / Control	Action
B	Draw a new bounding box
Mouse Wheel	Scroll through CT slices
Middle Click + Drag	Pan view
Right Click + Drag	Zoom in / out
Click row in box table	Centers all slice views on that lesion
Show checkbox (box table)	Hide / show that bounding box in the views; hidden boxes are still saved
Skip completed (checkbox)	Checked by default. Uncheck to browse or edit past cases
Slice thickness buttons	Switch 3 mm / TS / other; COR/SAG reconstruct from axial if needed
Finding type	PM nodule (R) default; also ascites, omental cake, stranding, lymph node